

Logger summary

DigiSolo.log

Project: Oeschibach | Serial: 453038986

Device: IGU-16HR 3C 5Hz | Firmware: V1.1.2.0be

File size: 17,699,484 bytes | Sections: 85,813 | Encoding: utf-8-sig

Timed-record UTC coverage: 2026-05-13T15:12:39+00:00 to 2026-09-11T04:53:49+00:00

Elapsed span: 120.570 days. Coverage describes logged observations and does not prove continuous waveform acquisition.

Section inventory

Family	Records	Contents
DeviceInfo	2	Boot/configuration, channel self-tests and storage identity
BatteryPowerOnStop	1	Power-stop marker; meaning not decoded
Notify	31	Acquisition start and acquisition file changes
GPS	59,134	Power-cycle, synchronization, location, satellites and orientation
Temperature	17,118	Temperature observations
Battery	5,759	Battery voltage observations
Memory	3,768	Total, available and used storage counters

Observations to review

No Error sections found. This does not establish that the device had no faults.

Preamble/header retained verbatim; its numeric tuple is not decoded without a format specification.

Boot RTC is invalid or in a different year from timed observations; excluded from coverage dates.

Some GPS records report Leap Second = 0; UTC fields are used as logged, without automatic correction.

Measurement statistics and sampling

Statistics use valid numeric values with valid record UTC times. First/last follow chronological order. Units are logged or explicitly assumed.

Metric / unit	n	Min	Median	Max	First / last
Battery.Voltage (V (assumed))	5,759	7.758	8.026	8.902	8.902 / 7.776
Temperature.Temperature (deg C (assumed))	17,118	2.75	14.12	26.69	21.19 / 10.88
Memory.Used Memory (logged units)	3,768	22	2.29e+04	4.58e+04	22 / 4.58e+04
Memory.Available Memory (logged units)	3,768	7.519e+04	9.808e+04	1.21e+05	1.21e+05 / 7.519e+04
GPS.Satellite Number (count)	21,325	3	6	10	4 / 5
GPS.Phase Error (unspecified)	18,905	0	27	204	1 / 11
GPS.Altitude (m (assumed; datum unknown))	21,325	1,054	1,392	2,227	1,397 / 1,387
GPS.eCompass North (deg (assumed))	21,325	40.08	53.83	56.2	55.05 / 40.08
GPS.Tilted Angle (deg (assumed))	21,325	6.574	8.836	10.2	7.043 / 6.727
GPS.Roll Angle (deg (assumed))	21,325	-8.644	-7.68	2.339	-6.362 / 2.339
GPS.Pitch Angle (deg (assumed))	21,325	-6.119	-4.353	6.304	-3.009 / 6.304

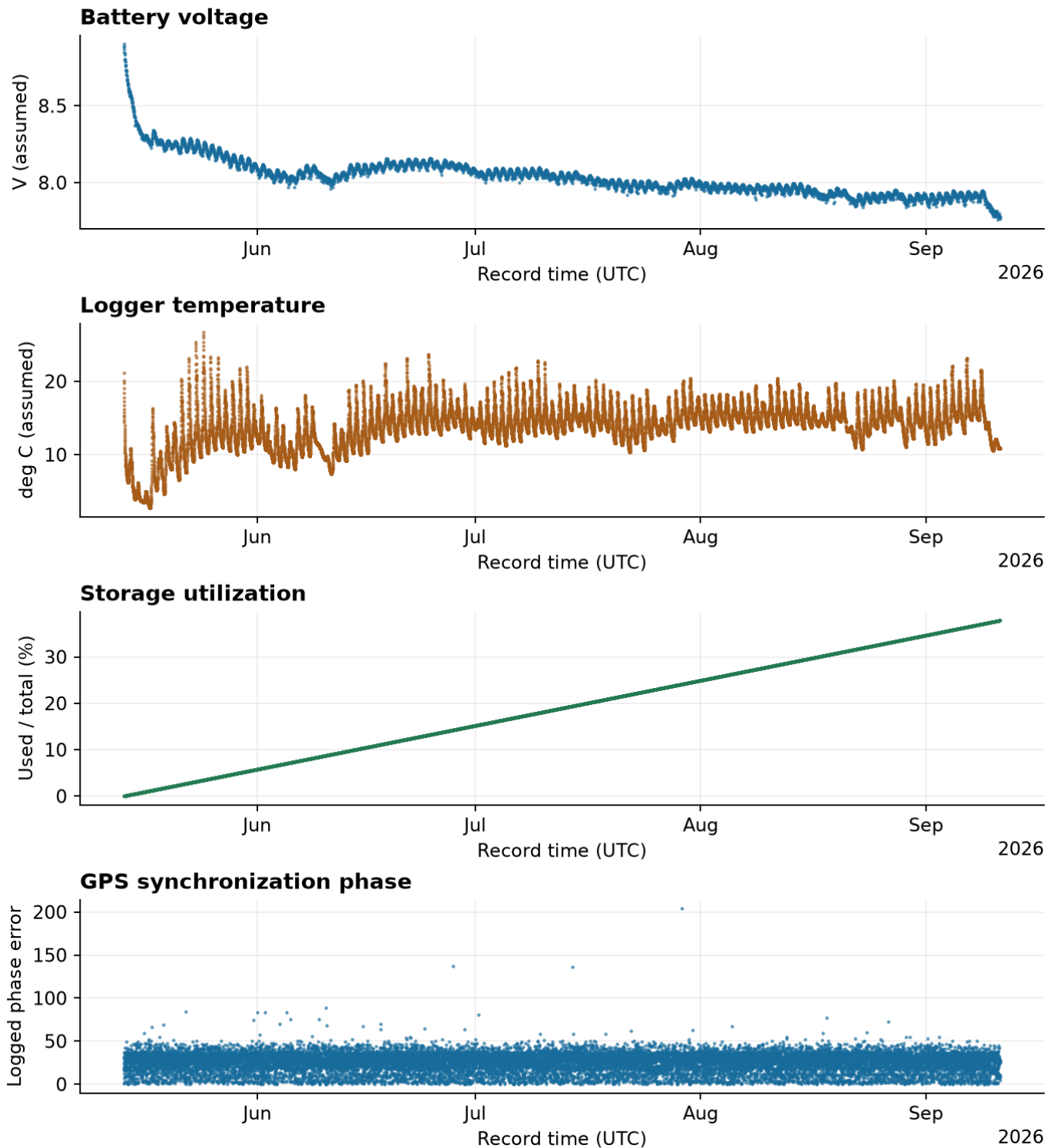
Field	Median interval (min)	Largest gap (min)	Gaps > 3x median
Battery.Voltage	30	30.4	0
Temperature.Temperature	10	10.4	0
Memory.Used Memory	46	52	0
GPS.Satellite Number	8	8.4	0

Cadence is the median positive interval for the named metric. Gap flags indicate unusually spaced logger observations; they do not establish gaps in DLD waveform data.

Last valid storage utilization: 37.85% of total. Absolute counter units are unspecified.

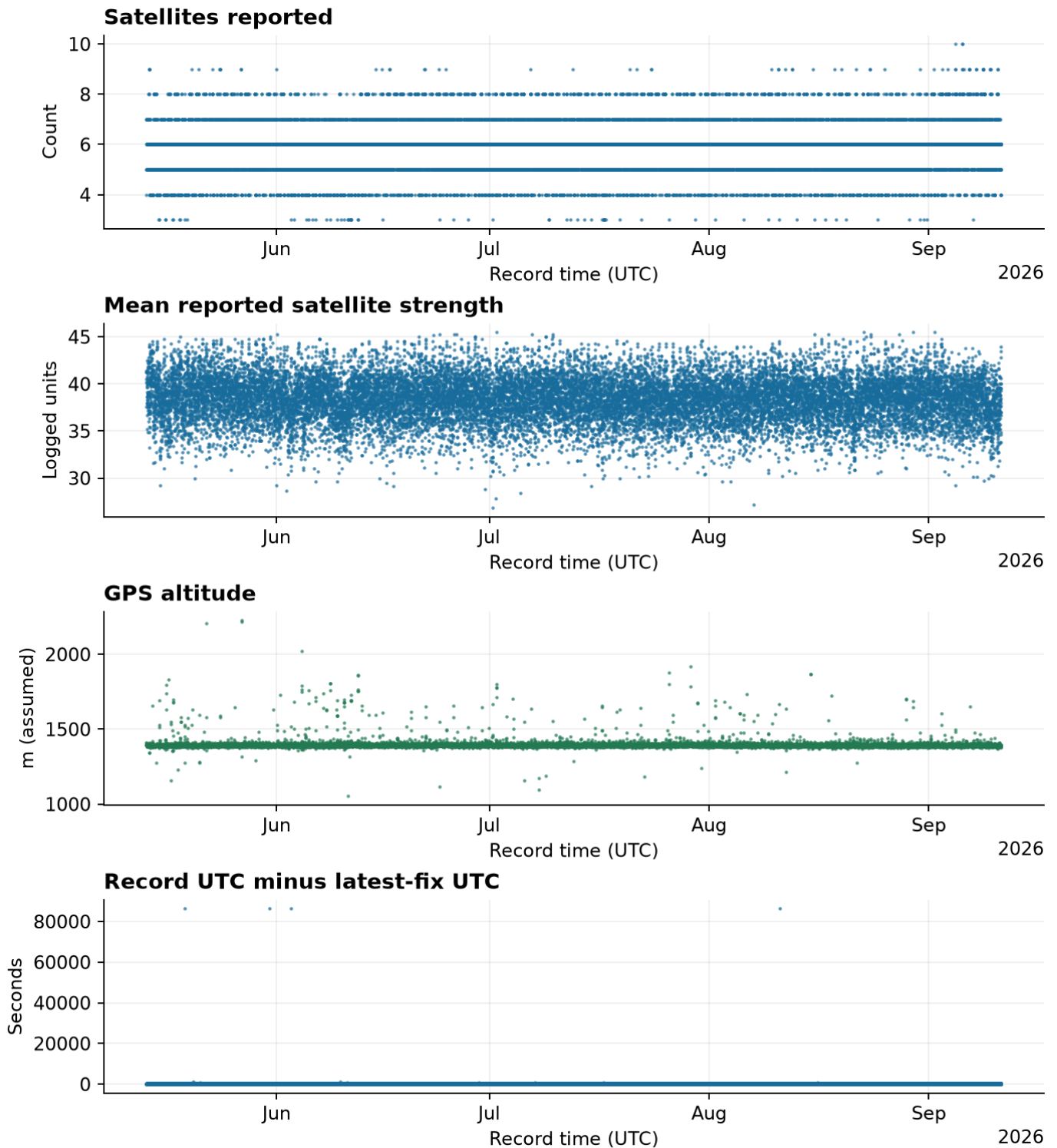
Power, temperature, storage and synchronization

All valid samples are shown as points. No interpolation or smoothing. Memory percentage avoids assuming counter units; phase-error units are unspecified.



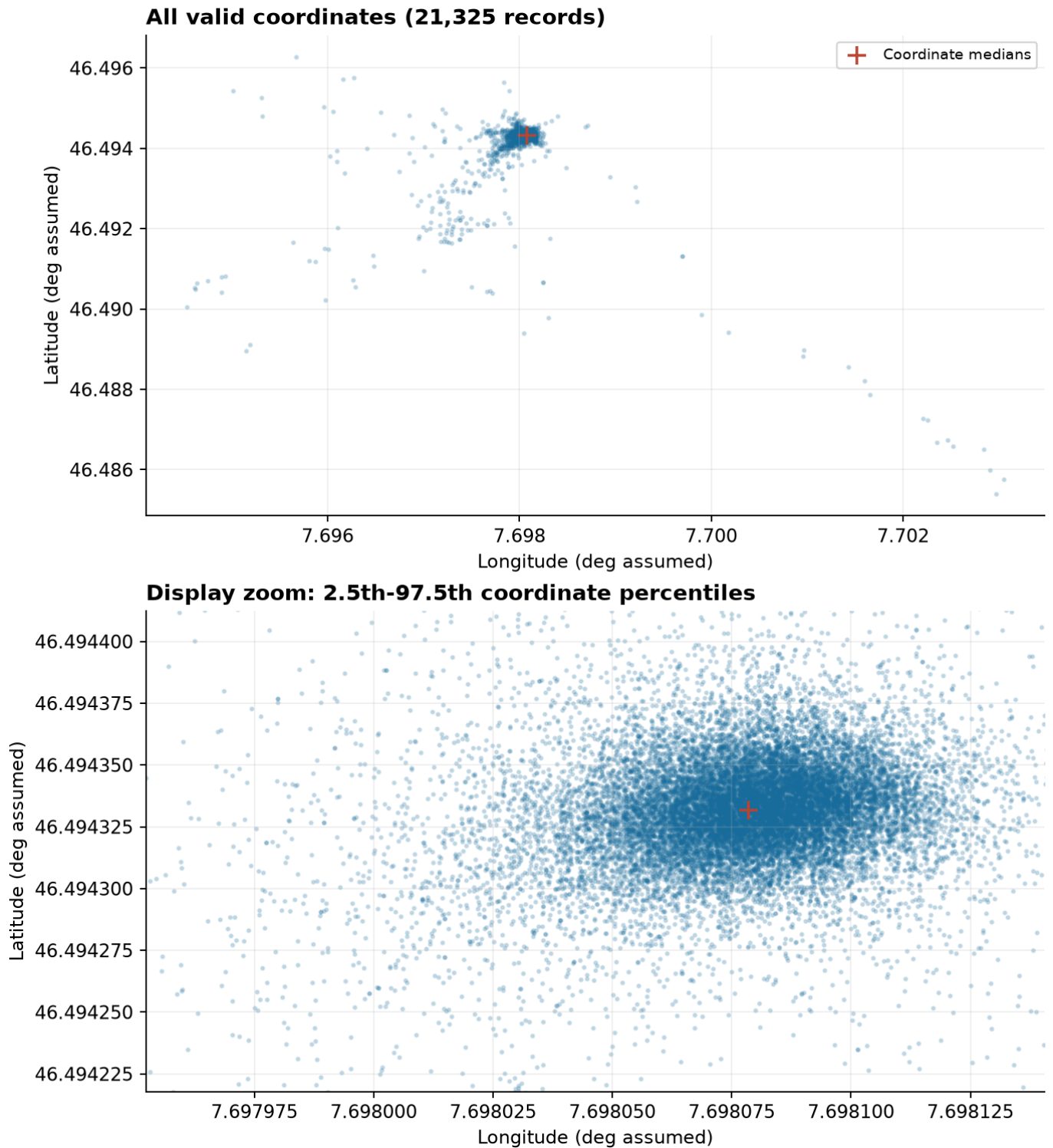
GPS observations and fix age

Satellite strength is the arithmetic mean of each numeric list. Fix age may be negative when timestamp fields describe different instants. No position outliers are removed.



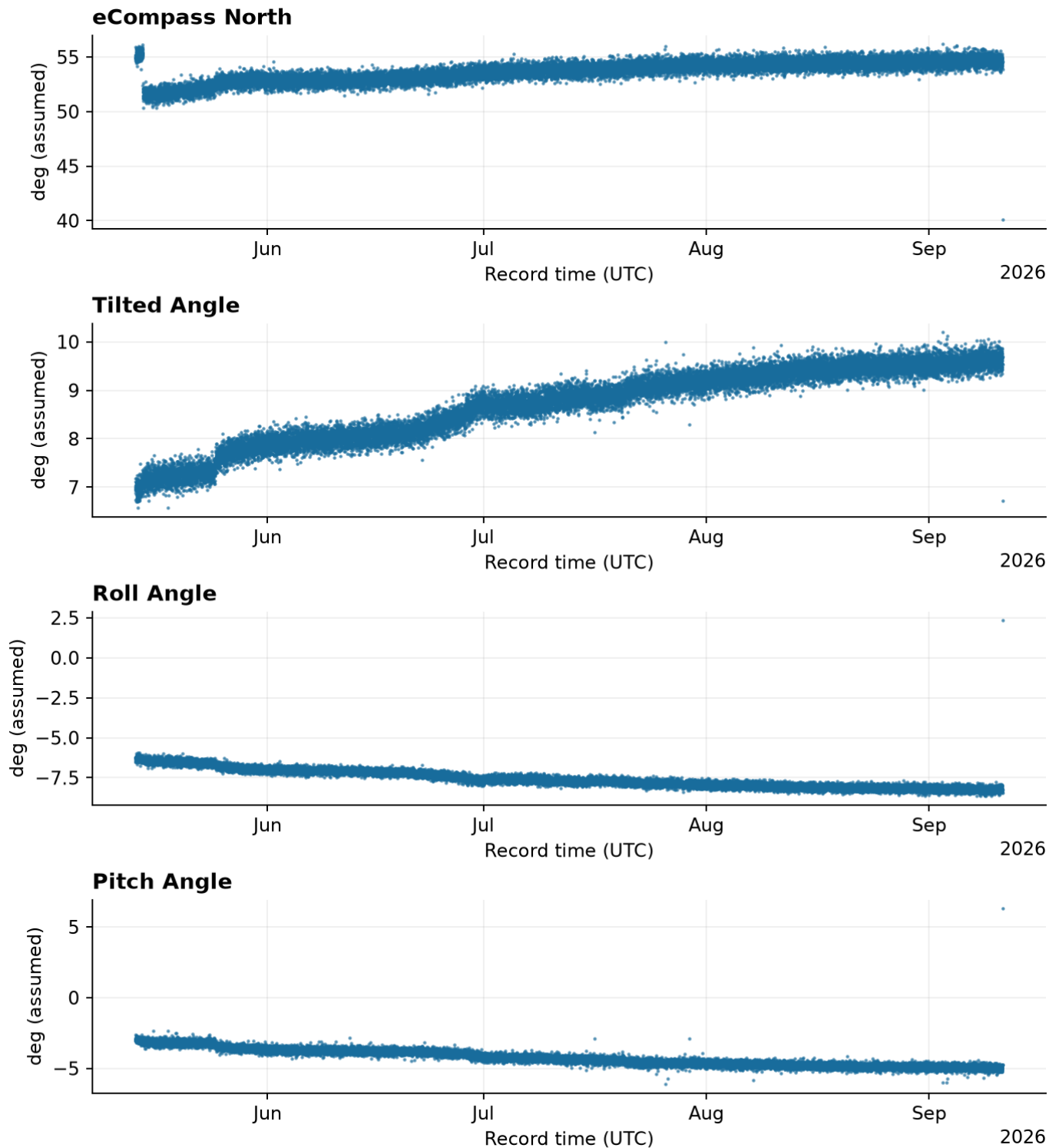
Reported GPS position scatter

The upper panel includes every coordinate in geographic bounds. The lower panel changes axis limits only. These are degree-coordinate plots, not metric maps; scatter is not proof of movement.



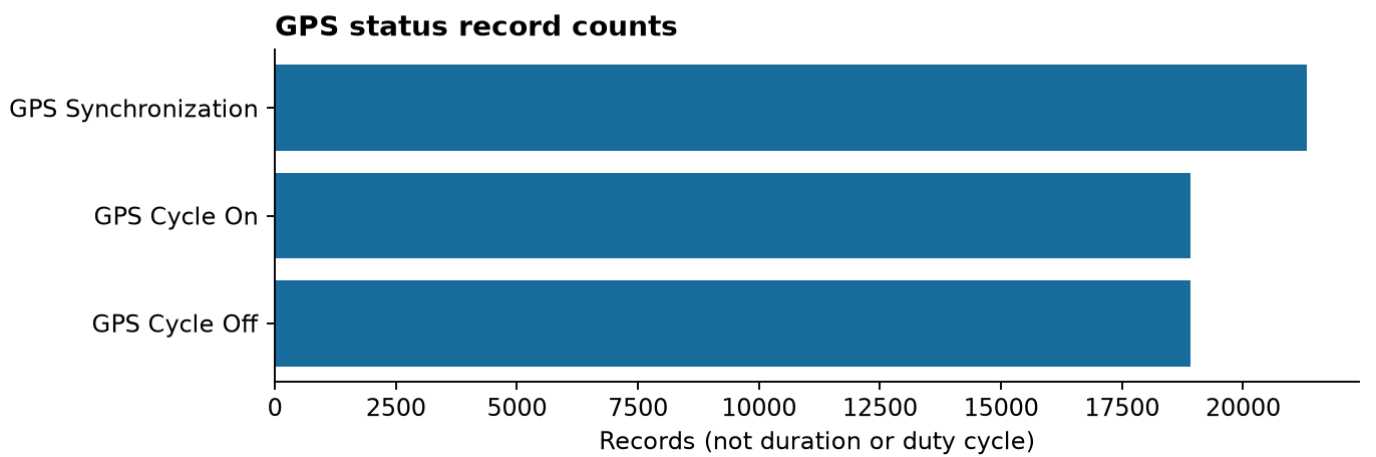
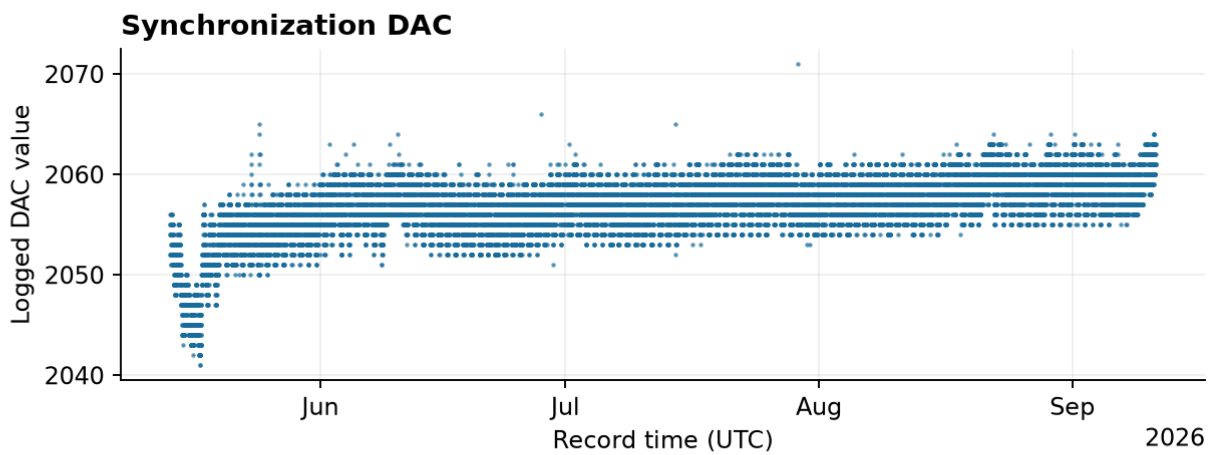
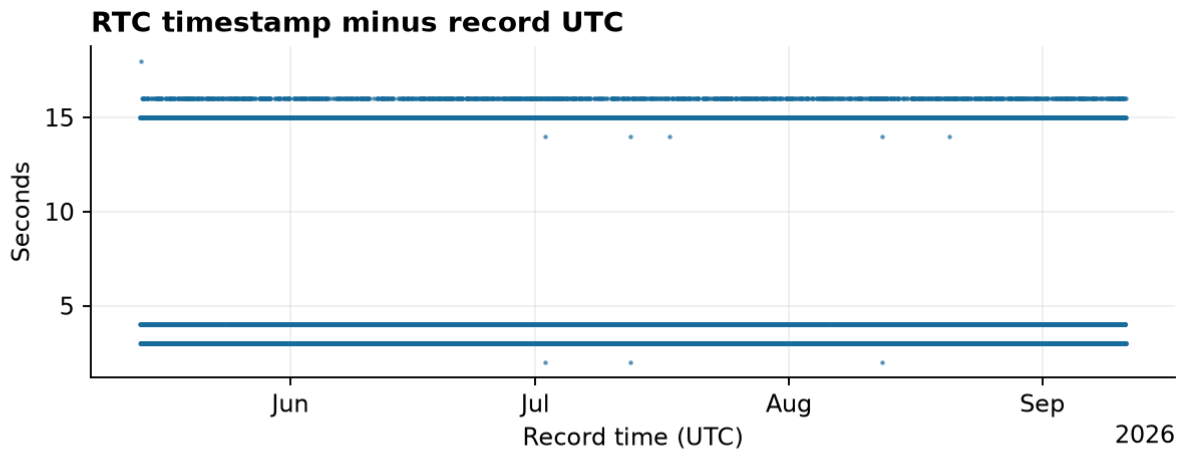
Compass and instrument orientation

Angles are plotted exactly as logged. Degrees are assumed; compass reference (magnetic or true north) is unspecified. No compass calibration or unwrapping is applied.



Timing fields and GPS operating states

RTC-minus-UTC is a comparison of timestamp fields, not a calibrated oscillator-error estimate. Status frequencies count records; they do not measure time spent in each state.



Boot configuration and channel checks

Each boot is kept separate. Boot reason codes, gains and test units are preserved without vendor-specific interpretation.

Field	DeviceInfo00001	DeviceInfo00002
BootReason	0007	0300
Boot RTC	2015/12/18,04:19:49	2015/12/25,04:10:54
GPS Lock Time	-	2026/05/13,15:12:08
Start Date	2026/05/06	2026/05/06
Sample Rate	200	200
Channel Number	3	3
GPS Power Mode	CycleOff	CycleOff
Voltage Threshold	650,600	650,600
Geophone Test Passed	-	1
Sensor Type	-	TMPx75

DeviceInfo00001

Channel	Resistance	Resonate Freq	Damping	Sensitivity	RMS Noise
1	1.7962, 1.8289	243.9827	0.4608	184.8908	1.2214
2	1.8033, 1.8437	4.9990	0.7095	79.3174	1.1769
3	1.7987, 1.8551	4.9479	0.7090	80.0688	1.1902

DeviceInfo00002

Channel	Resistance	Resonate Freq	Damping	Sensitivity	RMS Noise
1	1.7937, 1.8328	4.9376	0.6801	77.2093	1.2508
2	1.8031, 1.8374	4.9210	0.6948	77.4916	1.1685
3	1.8103, 1.8328	5.1128	0.6329	65.3755	1.1828

Complete device settings, including spread noise, SD identity and all raw pairs, are available in `parsed_log.json` and the `DeviceInfo` CSV.

Notifications and error log

All selected events in source order. Untimed markers retain their section and source line. Acquisition filenames are references; this report does not read the DLD files.

UTC / section / line	Details
No UTC BatteryPowerOnStop line 61	(empty marker)
2026-05-13T15:12:39+00:00 Notify00001 line 136	Start Acquisition FileName = Seis000.DLD
2026-05-17T15:12:37+00:00 Notify00002 line 25607	Changed Acquisition FileName = Seis001.DLD
2026-05-21T15:12:37+00:00 Notify00003 line 51089	Changed Acquisition FileName = Seis002.DLD
2026-05-25T15:12:37+00:00 Notify00004 line 76589	Changed Acquisition FileName = Seis003.DLD
2026-05-29T15:12:37+00:00 Notify00005 line 102064	Changed Acquisition FileName = Seis004.DLD
2026-06-02T15:12:37+00:00 Notify00006 line 127563	Changed Acquisition FileName = Seis005.DLD
2026-06-06T15:12:37+00:00 Notify00007 line 153037	Changed Acquisition FileName = Seis006.DLD
2026-06-10T15:12:37+00:00 Notify00008 line 178505	Changed Acquisition FileName = Seis007.DLD
2026-06-14T15:12:37+00:00 Notify00009 line 203983	Changed Acquisition FileName = Seis008.DLD
2026-06-18T15:12:37+00:00 Notify00010 line 229453	Changed Acquisition FileName = Seis009.DLD
2026-06-22T15:12:37+00:00 Notify00011 line 254929	Changed Acquisition FileName = Seis010.DLD
2026-06-26T15:12:37+00:00 Notify00012 line 280449	Changed Acquisition FileName = Seis011.DLD
2026-06-30T15:12:37+00:00 Notify00013 line 305913	Changed Acquisition FileName = Seis012.DLD
2026-07-04T15:12:37+00:00 Notify00014 line 331429	Changed Acquisition FileName = Seis013.DLD
2026-07-08T15:12:37+00:00 Notify00015 line 356939	Changed Acquisition FileName = Seis014.DLD
2026-07-12T15:12:37+00:00 Notify00016 line 382463	Changed Acquisition FileName = Seis015.DLD
2026-07-16T15:12:37+00:00 Notify00017 line 407941	Changed Acquisition FileName = Seis016.DLD
2026-07-20T15:12:37+00:00 Notify00018 line 433419	Changed Acquisition FileName = Seis017.DLD
2026-07-24T15:12:37+00:00 Notify00019 line 458897	Changed Acquisition FileName = Seis018.DLD
2026-07-28T15:12:37+00:00 Notify00020 line 484367	Changed Acquisition FileName = Seis019.DLD
2026-08-01T15:12:37+00:00 Notify00021 line 509883	Changed Acquisition FileName = Seis020.DLD

UTC / section / line	Details
2026-08-05T15:12:37+00:00 Notify00022 line 535374	Changed Acquisition FileName = Seis021.DLD
2026-08-09T15:12:37+00:00 Notify00023 line 560845	Changed Acquisition FileName = Seis022.DLD
2026-08-13T15:12:37+00:00 Notify00024 line 586309	Changed Acquisition FileName = Seis023.DLD
2026-08-17T15:12:37+00:00 Notify00025 line 611829	Changed Acquisition FileName = Seis024.DLD
2026-08-21T15:12:37+00:00 Notify00026 line 637319	Changed Acquisition FileName = Seis025.DLD
2026-08-25T15:12:37+00:00 Notify00027 line 662785	Changed Acquisition FileName = Seis026.DLD
2026-08-29T15:12:37+00:00 Notify00028 line 688316	Changed Acquisition FileName = Seis027.DLD
2026-09-02T15:12:37+00:00 Notify00029 line 713799	Changed Acquisition FileName = Seis028.DLD
2026-09-06T15:12:37+00:00 Notify00030 line 739281	Changed Acquisition FileName = Seis029.DLD
2026-09-10T15:12:37+00:00 Notify00031 line 764801	Changed Acquisition FileName = Seis030.DLD

Interpretation and reuse

Use record UTC Time for coverage and chronological plots; preserve other clocks separately.

Group headers by trailing numeric suffix; retain every original section and source line.

Preserve outliers and unknown fields. No automatic filtering, interpolation or unit conversion.

Temperature deg C, voltage V, angles deg, coordinates deg and altitude m are conventional assumptions, not declared units.

Memory is in unspecified logged units; use $\text{Used/Total} \times 100$ for storage percentage.

Phase Error, GPS Tick, gains and diagnostic units are not decoded. Voltage Threshold is retained raw.

Coordinate scatter describes reported fixes, not confirmed movement; zoom is display-only.

RTC minus record UTC and latest-fix age compare logged timestamps; neither alone proves a clock fault.

A gap over three times median positive cadence is a review flag, not proof of lost acquisition.

Log end is an observation boundary, not evidence of shutdown. DLD waveforms are separate files.

Reproduce this report

```
python digisolo_report.py "DigiSolo.log"
```

Use `--format plots` for PNG figures only, `--parse-only` for dependency-free JSON/CSV export, `--output-dir` to choose a destination, and `--encoding` for a known text encoding.

Use `--voltage-threshold` only for a threshold you explicitly choose. The raw Voltage Threshold field is not automatically scaled or interpreted.

For similarly structured logs, new section families and fields are retained. Timed scalar fields in unfamiliar families receive additional plots. Logs with a different grammar need a parser adapter.

Companion outputs: `summary.json` contains counts, statistics and warnings; `parsed_log.json` preserves key/value pairs and unparsed lines; `csv/` contains one file per section family; `events.csv` contains event details; `figures/` contains reusable PNG plots.